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The relationship between nature exposures and attention restoration, as moderated by exposure duration: A systematic review and meta-analysis☆

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ABSTRACT

Much prior research suggests that exposure to natural environments is associated with improved cognitive performance, in comparison to non-natural exposures. However, the presence and size of such nature benefits are inconsistent. This meta-analysis aimed to determine the influence of nature exposure duration, a likely moderating factor, on cognitive performance change. In total, 273 individual outcomes from 80 studies were included in correlated hierarchical effects (CHE) multilevel meta-analyses. When analysing all cognitive outcomes, the average performance improvement was larger for nature exposures compared to non-natural comparators. Of 11 cognitive domains identified in the literature, Working Memory and Attentional Control were the only domains to show reliable nature benefits. However, subgroup analysis suggested that effect size differences between cognitive domains may not be significant. Meta-regressions indicated exposure duration significantly moderated natural/non-natural differences in restoration in a non-linear relationship, though the shape and significance of this relationship within each cognitive domain was not consistent. Nature benefits were generally larger for participants who underwent cognitive fatigue, but other experimental design variations did not produce meaningful differences in effects. Overall, effects sizes were small and heterogeneity was substantial. The largest difference in cognitive restoration between natural/non-natural exposures appears to occur after approximately 30 min of environmental exposure.

1. Introduction

1.1. Background

The ability to remain focused on specific tasks for extended periods is supported by selective or *directed attention*, a cognitive resource described by Attention Restoration Theory (ART; Kaplan, 1995) as the conscious, volitional focus on salient stimuli and the simultaneous engagement of inhibitory control processes that supress distracting stimuli.

Continued application of directed attention can result in attentional fatigue and consequent declines in performance on tasks that are dependent on this cognitive resource. The central claim of ART is that natural environments produce greater opportunity for recovery from attentional fatigue. This is largely attributed to natural environments purportedly eliciting an alternative, bottom-up form of attentional activation (soft fascination) to disengage the top-down directed attention, along with nature satisfying other key elements of restoration such as Being Away, Extent, and Compatibility (see Kaplan, 1995 for review). The result being improved attention restoration in nature, relative to non-natural settings such as built/urban or indoor environments.

Extensive research has investigated the effect of environmental exposures on cognition. While some studies have reported strong benefits of nature exposures on attentional outcomes relative to non-natural exposures (Berman et al., 2008; Berto, 2005), null differences are also commonly reported (Hicks et al., 2020; Johnson et al., 2021). An early systematic review and meta-analysis by (Bowler et al., 2010) concluded there was a moderate effect favouring natural settings, however, given only five effect sizes were included in the analysis (mixing subjective and objective attentional measures) this finding was largely exploratory.

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According to ART, 'bottom up', involuntary attentional processes (Corbetta & Shulman, 2002; Fan et al., 2005) are less dependent on directed attention, and thus task performance may be largely unaffected by directed attention fatigue. Conversely, higher order tasks that depend on (top-down) directed attention will be differentially impacted by attentional fatigue, meaning these tasks have more potential for between-group (nature vs non-nature) restorative differences to emerge than tasks where performance is driven by bottom-up processes (Berman et al., 2008).

Ohly et al. (2016) performed a meta-analysis of the results of 31 ART studies using objective cognitive outcomes. They pooled the effects of environmental interventions on performance independently for different standardised cognitive tests, enabling investigation of whether performance on certain tasks is more sensitive to restoration in nature than others. While nature had a positive effect on Digit Span Forward/Backward Test (DSF/B) and the Trail Making Test Part B (TMTB) performance, non-natural control settings had a positive effect on the Search and Memory Task. Performance from all other tasks did not differ between environmental settings.

Stevenson et al. (2018) built on Ohly et al.'s (2016) approach by grouping cognitive outcomes from 46 studies into different 'cognitive domains', based on similar task characteristics or underlying attentional/cognitive processes the task purported to measure. For example, where Ohly et al. distinguished effects for the Necker Cube Pattern Control Test (NCPCT) and conflict scores on the Attention Networks Test (ANT), Stevenson et al. pooled the effects from these tasks in a domain labelled 'Attentional Control'. In total, Stevenson et al. identified seven cognitive domains including Attentional Control, Cognitive Flexibility, Impulse Control, Processing Speed, Vigilance, Visual Attention, and Working Memory. Small-to-moderate standardised mean differences (SMD) favouring natural settings were found for Working Memory, Cognitive Flexibility and Attentional Control. In further subgroup analyses, actual nature was associated with larger restorative effects than virtual nature exposures, and effects in some domains were larger when there was greater restoration potential (i.e., an initial fatigued status).

More recent meta-analyses of singular task outcomes support the notion that working memory (DSB performance) improves with nature exposures (Stenfors et al., 2019), though nature benefits are less clear on executive network outcomes of the Attention Networks Test (Johnson et al., 2021). While these findings align with those made by Stevenson et al. (2018), no statistical contrast has been conducted to quantify the presence/absence of nature restoration benefits for different cognitive domains. Rather, differences in statistical significance between independent meta-analyses have been used to evidence restorative effects existing in a select few cognitive domains. This can be inappropriate given the inherent reliance on statistical power (Borenstein et al., 2021; Harrer et al., 2021) which can differ greatly across cognitive domains. For example, working memory research is prevalent in ART literature, likely resulting in comparatively highly powered meta-analyses relative to other domains (e.g., creativity). Claims that pooled SMDs meaningfully differ between cognitive domains can be better supported by statistically quantifying those differences through formal subgroup analysis, rather than informal comparison of results generated by independent meta-analyses of each domain.

1.2. The current review

Building and extending upon previous systematic review and meta-analysis of ART literature, this review intends to provide novel synthesis and analysis of the quantitative evidence regarding the effect of natural environments on cognition. Adopting Stevenson et al.'s (2018) strategy of grouping cognitive outcomes into domains, this review uses contemporary statistical analyses to investigate previously unexplored variables and provide an updated review of the broad and burgeoning ART literature.

The primary aim of the current review was to investigate exposure duration as a potential moderator. The duration of environmental exposures in studies ranges widely, from short 'micro' exposures lasting under a minute (Johnson et al., 2022; Lee et al., 2015), to 'moderate' length exposures under an hour (e.g., Amicone et al., 2018; Berman et al., 2008; Wade et al., 2020), to longer exposures of several hours (Triguero-Mas et al., 2017). In prior meta-analyses, the effects associated with these varying intervention durations have been pooled together without further exploration. Given dose may be a crucial factor for intervention efficacy, empirically clarifying the relationship between exposure duration and effect size is essential, particularly for the development of evidence-based guidelines and practice recommendations for nature exposure treatments.

In this review, we operationalise duration of exposure (minutes) as the 'dose' of acute environmental interventions, and speculate that restorative effects in each environmental setting are likely to follow a dose-response curve – though presumably with differing rates of response (faster in nature; e.g., Fig. 1a). Such a relationship would indeed predict the previously observed pooled effect sizes favouring nature (e.g., Stenfors et al., 2019; Stevenson et al., 2018), though based on the restorative logic of ART (i.e., restoring a resource to baseline) these differences cannot be maintained linearly over extended exposure durations. That is, nature cannot perpetually improve cognition to a greater degree than non-natural settings when the end result is homeostasis, which will also eventually be reached under any conditions where directed attention is suitably disengaged (Kaplan, 1995). Instead, we predict a non-linear relationship between exposure durations and between-group SMDs should emerge (approximating the shape in Fig. 1b), which is tested in the current review via quadratic meta-regression. However, it is possible that nature benefits not strictly due to directed attention restoration to a baseline level, but involve enhancement of some other interrelated function to improve cognition (i.e., 'instorative' effects; Gidlow et al., 2016; Hartig, 2007). Compelling linear relationships between exposure duration and effect size could plausibly support instorative perspectives.

The current review also differs from prior research in analytic strategy regarding pre-post intervention data to assess restoration. Previous meta-analyses all make an effort to consider pre-intervention performance, whether by basic post/pre subtraction of SMDs (Bowler et al., 2010), or categorising studies as 'baseline balanced' (SMD <0.1 at pre-test) and then implementing sensitivity (Ohly et al., 2016), or by subgroup analyses (Stevenson et al., 2018). Despite these methods, each publication still makes inferences about performance *change* based on the analysis of post-test data only. Furthermore, these strategies involve removing valid, randomised, pre-post experimental data from analyses. To retain maximal data from the literature while ensuring outcomes are not biased by pre-intervention differences, we analyse pre-to-post-intervention change scores, contrasting performance improvements between natural and non-natural settings. This shifts the analysis to metrics that reflect the effect of the intervention, with a null hypothesis that each environmental intervention produces an equal *change* in performance.

Lastly, there is substantial methodological heterogeneity in the ART literature. Interventions can be either 'real' (actual) or virtual, with each of these able to be either sedentary or active. If active, the degree of activity can range from gentle walking to vigorous exercise. For virtual exposures, the mode and immersion of the exposure varies, including still images only, audio only, images and audio, video only, video with audio, and so on. On top of these variations, primary studies are commonly not only interested in a general environmental effect, and include other experimental manipulations to investigate interactions. These can be physical adaptations to the exposures, such as completing them with friends, phones, or laptops present. Alternatively, manipulations can target relevant non-physical constructs, such as social identity (Morten et al., 2017). Such factors likely moderate environmental effects, as previously seen with actual vs virtual exposures (Stevenson et al., 2018). Therefore, the current review will investigate moderating

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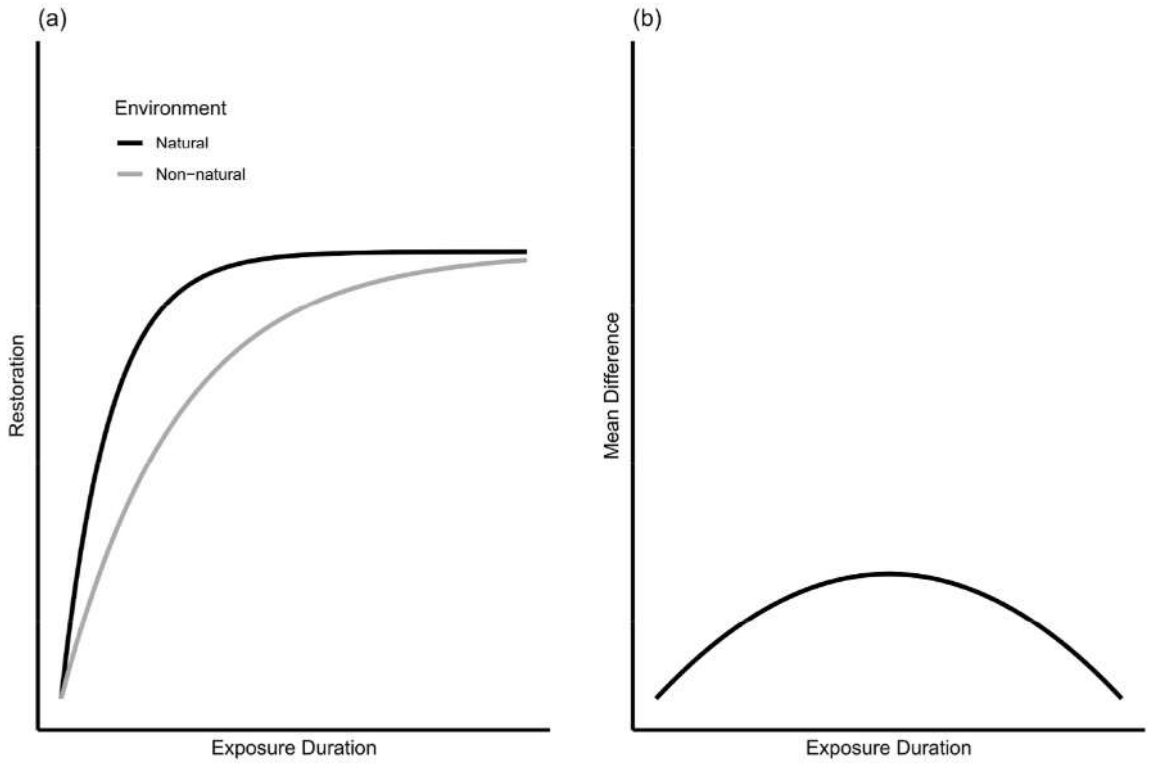


Fig. 1. The Hypothetical effect of time on restoration in natural and non-natural settings (a), and (b) an approximate corresponding Hypothetical relationship between time and the mean difference in restoration of those groups.

factors via subgroup analysis to determine if these methodological differences in experimental design are associated with differences in pooled effect sizes.

1.3. Objectives and hypotheses

Objective 1: Estimate the effect of nature interventions on pre-versus-post cognitive performance change, relative to that of non-natural interventions, for each distinct cognitive domain. In addition, statistically test the difference between within-domain effect sizes to add validity to conclusions regarding presence/absence of effects in given domains. The pooled SMD was predicted to be larger in cognitive domains which theoretically rely more heavily on directed attention (H_1), such as Working Memory, Attentional Control and Cognitive Flexibility (Stevenson et al., 2018).

Objective 2: Within each cognitive domain, estimate the predictive utility of exposure duration on the average SMD of performance improvement between natural and non-natural exposures and determine what shape best describes this dose-response relationship. A quadratic relationship between exposure duration and SMD was hypothesised.

Objective 3: Within each cognitive domain, exploratory subgroup analyses will estimate the impact of methodological sources of heterogeneity in predicting the average SMD between natural and non-natural exposures.

2. Materials and method

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines and Cochrane recommendations (Higgins et al., 2023). The review protocol was registered with PROSPERO in August 2022 (registration ID: CRD42022352330).

2.1. Literature search & selection process

A systematic search strategy was devised by the research team with the assistance of independent research librarians from the University of Tasmania. The goal of the search was to capture any experiments examining changes in objective cognitive performance following exposure to natural environments or stimuli. Three concepts were used to systematically search the literature: *nature*, *cognition*, and *tests/tasks*. These three concepts mirror the search strategy employed previously by Stevenson et al. (2018).

The final list of search terms (supplementary material S1) was applied to search the titles, abstracts and keywords of articles in MEDLINE, PsycINFO, SCOPUS, Web of Science and EMBASE databases. The literature search is current up to August 24th, 2023. The database searches (an initial 2022 search and 2023 update) were downloaded to Endnote21 (Clarivate, 2013) and screened for duplicates, before being uploaded to the online software Covidence (Veritas Health Innovation, 2023) for screening. Additional relevant studies identified in the references of screened articles were manually included if appropriate. Title/abstracts of all search results, and subsequent eligible full texts, were each independently double-screened by the review team. CB (corresponding author) screened all titles/abstracts and full texts, with second screening undertaken by one of the other authors. Any disagreements between reviewers were explored and resolved during resolution meetings. Details of disagreements are not reported nor statistically quantified, in accordance with Cochrane guidelines (Higgins et al., 2023).

2.2. Eligibility criteria

Table 1 contains the eligibility criteria applied throughout screening. The criteria aimed to facilitate the collection of rigorous experimental studies.

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Table 1
Eligibility criteria applied during Title/Abstract and full-text screening.

a) Human participants
b) Written in English
c) Peer reviewed journal article
d) Experimental intervention involving nature or natural stimuli
e) Comparator intervention with non-natural exposure
f) Objective behavioural test/measure of cognition
g) Pre- and post-test of cognitive measure
h) Random allocation to experimental and control conditions, or counterbalanced allocation in repeated measures designs.

2.2.1. Population

All study populations were eligible, regardless of age or clinical condition.

2.2.2. Intervention & comparator

Included studies implemented a mixed within-between experimental design, where an experimental and comparator group were measured pre-and-post intervention. Allocation to experimental/comparator groups must have been randomised (individual or cluster). Experimental group interventions must have included the presentation of 'natural' stimuli, whether fully immersive or instances of nature added to otherwise control settings. Comparator group interventions must have been (predominantly) void of 'natural' stimuli, either as passive control groups or as alternative exposure groups to urban/built stimuli.

Given a focus of the review was to assess the influence of exposure duration, only results from one-off acute interventions were synthesised, rather than cumulative longitudinal programs. This distinction was not applied during screening, with ineligible longitudinal designs identified during data extraction.

2.2.3. Outcomes

Included studies needed to report pre- and post-test data from at least one objective, behavioural measure of cognitive performance.

2.3. Data extraction

A standardised data extraction spreadsheet was developed by the research team, based upon the Cochrane (Higgins et al., 2023) and Covidence templates. Study authors were contacted where information was unclear or unattainable from the publication. Data extraction was carried out by CB. Pre- and post-test means (*M*) and standard deviations (*SD*) were extracted on all normally distributed cognitive outcome measures. Crucially, change-*M* and change-*SD* (required to generate SMDs of performance change following interventions) were extracted or calculated. Where change-*SDs* were not reported, unable to be obtained via author follow-up, or unable to be calculated via summary statistics, these values were imputed using the recommended methods outlined in the Cochrane Handbook (Higgins et al., 2023). The source of Change-SD values (True vs Imputed) was included in subgroup analyses.

2.4. Quality appraisal & risk of bias

The quality appraisal (QA) system was an adaptation of a field-specific system designed and used by Ohly et al. (2016) and Stevenson et al. (2018). It combines key indicators and items from three independent review tools to provide a context-appropriate assessment of evidence in its degree of robustness and potential for bias. The QA checklist in the current review comprises 19 items, scored as 'yes' (2), 'partial' (1), 'no' (0) or 'unclear' (0), with total scores out of 38 expressed as a percentage. When scoring of QA items in primary studies were unclear, authors were contacted directly for clarification. See supplementary material S2 for quality appraisal checklists for eligible studies.

The Cochrane Risk of Bias 2 (RoB2; Sterne et al., 2019) instrument was used to identify risk of bias in randomised trials across the key domains of randomisation processes, deviations from intended interventions, missing outcome data, measurement of outcomes, and selection of reported results. The RoB2 allows for varying bias assessments for different outcome measures within a single study, with a judgement algorithmically generated for each domain along with an overall RoB for the outcome. All QA and RoB assessments were undertaken by CB. See supplementary material S3 for the RoB profile of all included outcome measures.

2.5. Data synthesis and analysis

Outcome measures were categorised according to overarching cognitive domain as deemed by the research team. The set of cognitive domains established by Stevenson et al. (2018) was used as an initial framework and expanded upon. A detailed rationale for cognitive domain classification for each cognitive task/outcome is presented in supplementary material S4.

Correlated and Hierarchical Effects (CHE) three-level meta-analyses were planned to allow for the inclusion of dependent observations and retain maximal nuance in the extracted data. This approach limits reviewer-imposed common-denominator decisions regarding data inclusion (Hartig & Jahncke, 2017), however, some combining of data was still required to avoid 'unit of analysis' issues (Higgins et al., 2023). Following Cochrane recommended practice, we combined intervention conditions where necessary to avoid double counting, unless this would limit the ability to run subgroup analyses related to comparator conditions (urban vs control) or type of exposure (actual vs virtual), in which case the *n* of each condition was manually divided by the number of times that condition appears.

Cluster randomised data were integrated using effective sample size adjustments via design effects, following Cochrane guidelines, using an assumed IntraCluster Correlation (ICC) value of 0.03. ICC values of <0.05 are typical and increase as cluster sizes decrease (Higgins et al., 2019). We included sensitivity analyses where cluster randomised *n* were adjusted with varying ICC values between 0 and 0.1 (supplementary material S5).

Once all contrasts were finalised, SMD effect sizes, corrected for small samples (Hedge's *g*), were calculated using the Change-*M* and Change-*SD* of each natural vs non-natural contrast identified in the eligible studies. All outcomes were coded such that a positive value indicated an improvement on the outcome. Positive between-group SMD values indicated an effect favouring nature interventions, with interpretation guided by Cohen's (2013) guidelines (0.20 = small, 0.50 = moderate, 0.80 = large). All meta-analyses were undertaken using the metafor package in R (Viechtbauer, 2010).

2.5.1. Objective 1 – pre-post intervention effects within cognitive domains

SMDs were pooled within each cognitive domain using independent CHE three-level meta-analyses with Robust Variance Estimation (RVE). The constant sampling variation was assumed to be 0.6. Sensitivity analyses for varying correlation values are plotted in supplementary material S6. CHE-RVE multilevel models not only account for the non-independence between multiple effects from a single study, but explicitly model a correlation structure of effect sizes within each study and protect against model misspecification and heteroscedasticity in

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coefficient estimates (Harrer et al., 2021; Pustejovsky & Tipton, 2022). The significance of effects in each domain cannot be validly contrasted to each other, as these domain analyses essentially represent observational subgroup analyses (Higgins et al., 2023). Formal, statistical contrast of within-domain effects occurs via subgroup analysis treating each domain as a subgroup within the broader dataset of cognitive outcomes.

A Subgroup Correlated Effects (SCE) RVE three-level model, with an assumed constant sampling correlation of 0.6, was undertaken to statistically test the hypothesis that average SMDs differ between the identified cognitive domains. SCE models are akin to CHE models within subgroups, but treat nested effects independently between subgroups (Pustejovsky & Tipton, 2022). This allows for the statistical comparison of subgroup effect estimations exactly as they were estimated independently. In essence, after fitting each independent cognitive domain CHE meta-analysis, the SCE model can quantify the difference between those pooled SMDs.

2.5.2. Objective 2 – moderating effect of exposure duration

Meta-regressions were undertaken by adding exposure duration as a covariate to each within-domain CHE-RVE model. Linear and quadratic terms of Exposure Duration were included allowing model fit comparison (Akaike Information Criterion; AIC) of linear and non-linear relationships between exposure duration and between-group SMDs. Following Cochrane recommendations (Higgins et al., 2023), meta-regressions were only undertaken in domains with 10+ observations.

2.5.3. Objective 3 – subgroup analyses of hypothesised points of heterogeneity

Lastly, individual subgroup variables are added to each domain's unmoderated model to assess for differences in estimated effects. SMDs incorporating true change statistics were first contrasted against SMDs based on imputed change-*SD*s, to determine if the imputation method biased the estimates. The following subgroups were then contrasted via independent inclusion in the model: Type of nature exposure (Actual/Virtual), mode of nature exposure (Visual/Auditory/Immersive), Density of nature exposure (Nature scene/Control scene with added nature), Type of comparator (Urban/Control), Activity (Exercise/Walking/Sedentary), Experimental fatigue induction (Yes/No), Other concurrent manipulation (Yes/No), Single Cognitive task (Single/Battery), sample age demographic (Children/Teens/Adults/Older-adults), clinical sample (Yes/No). Following Cochrane recommendations (Higgins et al., 2023), subgroup analyses were only undertaken in domains with 10+ observations. If a subgroup had <2 observations restricting pooling of effects, the subgroup was merged to retain the data in the analysis (e.g., Walking and Exercise exposures combined as an 'Active' subgroup).

2.5.4. Heterogeneity estimates

Heterogeneity is reported for each fitted CHE model via two statistics. I^2 is reported in accordance with Cochrane conventions (Higgins et al., 2023) to depict the proportion of overall heterogeneity attributable to variance in true effects, at both within-study (Level 2) and between-study (Level 3) levels. However, given I^2 does not portray the absolute heterogeneity of effects (Borenstein, 2019, 2024), we primarily convey heterogeneity using 95 % prediction intervals (PI) which depict an estimated dispersion window for true effects, relative to the pooled mean.

2.5.5. Multiplicity correction

Given the large number of significance tests, a study-wide correction for multiplicity was applied, with effects deemed statistically significant at an alpha of 0.01 (Borenstein et al., 2021; Polanin & Pigott, 2015; van Agteren et al., 2021), thus reducing the type I error rate to 1/100 tests across the entire publication.

2.5.6. Publication bias

Risk of small-sample bias (consistent with publication bias) was assessed through visual inspection of funnel plots and Egger's Regression Tests adapted for dependent data (i.e., Egger multilevel models; Rodgers & Pustejovsky, 2021). Where problematic funnel plot asymmetry and/or significant Egger's tests were observed, exploratory corrections were applied by deletion of outlier effects. In addition, cluster bootstrapped selection models (metaselection package; Pustejovsky et al., 2025) provide selective-reporting-corrected model parameter estimates, a novel method for assessing and handling publication bias-like reporting with dependent data. Results from these sensitivity analyses are reported in supplementary material S7.

3. Results

3.1. Part 1: systematic review

3.1.1. Search results & study characteristics

A total of 7056 records were identified from database and citation searching. Following screening, 86 articles, containing 93 individual studies/experiments, were deemed eligible for data extraction (see Fig. 2 for PRISMA flow diagram). These 93 studies included approximately[1] 6363 participants (61% women) across a range of child (<18 years old; $n = 912$), adult (≥18 years old $n = 5288$) and older adult samples (as defined by primary study authors; $n = 163$). Ten studies recruited from 'clinical' populations ($n = 130$ psychological conditions; $n = 377$ physical conditions). Mean sample age ranged from 6.8 to 86. Most studies recruited relatively young, university student participants (51%), with the median mean sample age being 22.1 years. A total of 23 countries contributed to this literature (Supplementary material S8a) with the earliest publication year being 1991 (Hartig et al., 1991), while 46 studies (~50%) were published in the previous five years from 2019 to August 2023 (Supplementary material S8b).

As outlined in section 2.2, four studies with longitudinal programs were excluded from the quantitative synthesis. A further nine studies were excluded during data extraction, primarily due to inability to obtain a relevant or normally distributed *M*, *SD*, or *n* (Fig. 2). Data from 80 studies (74 publications) were thus included in the final meta-analyses (Table 2). Of these, 50 (63%) were parallel trials, while 30 (37%) studies were crossover trials with washout periods ranging from 2-min breaks to multiple weeks. Participants in six studies were cluster randomised based on classroom/lecture membership (Amicone et al., 2018; Lin et al., 2014; Morton et al., 2017; Wade et al., 2020) or experiment date availability (Johnson et al., 2019; Schertz et al., 2022). Fig. 3 maps the evidence drawn from these 80 studies across several other key heterogeneity factors, including Age, Fatigue Induction, Exposure Duration, Actual/Virtual exposure, Urban/Control comparator setting, and Cognitive Domain.

Studies employed either Virtual (56%) or Actual (i.e., 'real'; 43%) environmental exposures. One study included an actual outdoor natural exposure and an indoor virtual exposure in the same experiment (Lee et al., 2022). Most designs contained immersive exposures (60%), of which 67% were actual exposures, while (33%) had virtual immersion through video recordings with audio, or still images presented with audio recordings. Visual-only exposures were observed in 31 studies and auditory-only exposures in three. Three studies contained more than one mode of exposure; Lee's (2022) actual vs virtual distinction also corresponded to an immersive/visual distinction. Additionally, Ma and Shu (2018) implemented visual and auditory-only exposures, as did Emfield and Neider (2014), although in the latter these were combined during

[1] We qualify these values as approximate due to sample sharing among studies (e.g., Evensen et al., 2015 & Raanaas et al., 2011; Han, 2017 & 2021) and instances where no sex/gender demographics or sample size are reported (Bernardo et al., 2021; Gamble et al., 2014; Sun et al., 2022; Wang et al., 2016).

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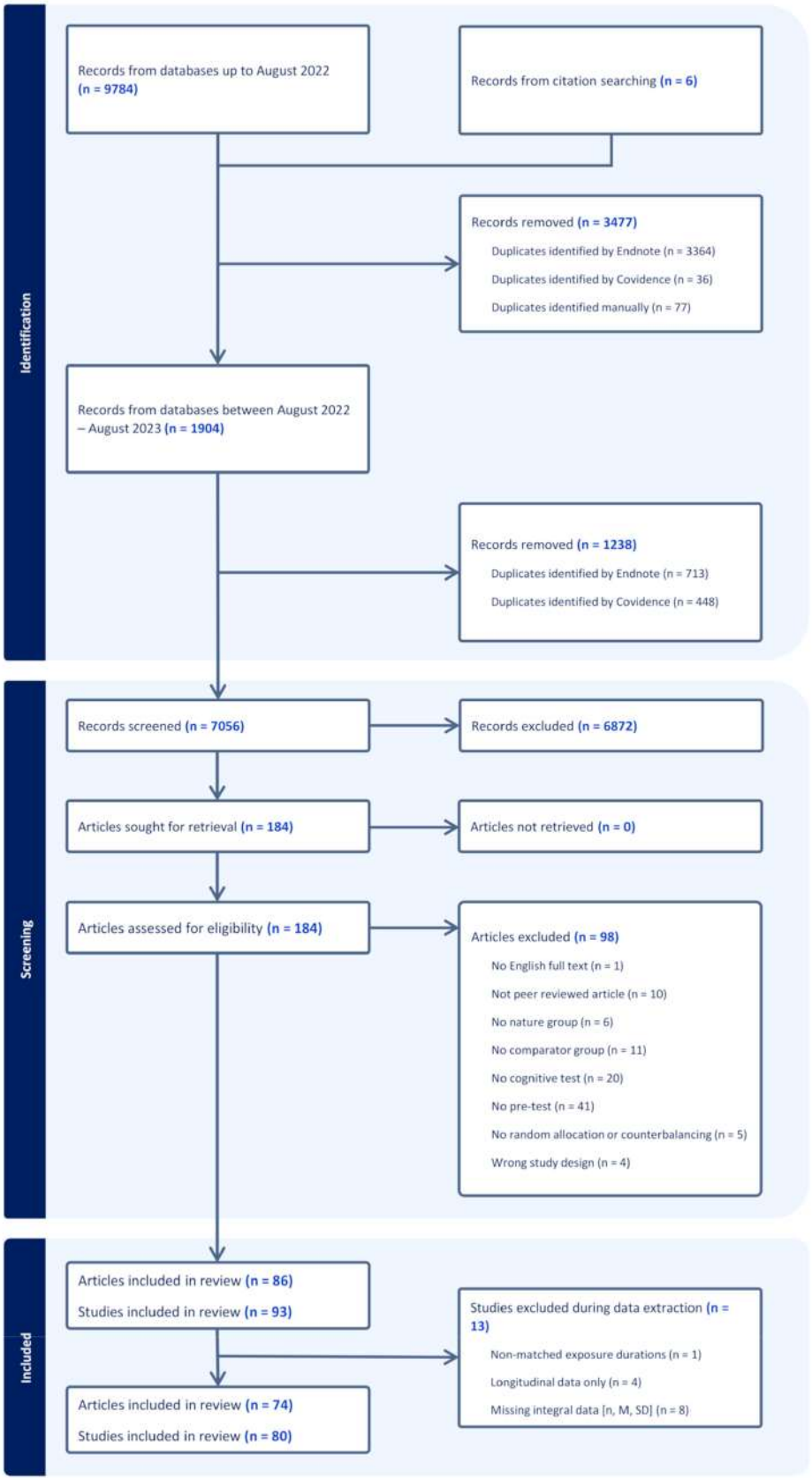


Fig. 2. Prisma flow chart of study selection process.

Table 2
Meta-analysed studies with cognitive task, outcome and Domain.

Author (Year)	Task	Outcome	Cognitive Domain
Amicone et al. (2018)	Bells Test	Total Correct	Visual Tracking
	Go/No-go Task	Total Correct	Inhibition
	Digit Span (Forward & Backward)	Combined Total	Working Memory
Anabitarte et al. (2021)	Attention Network Test (Children)	Test Score	Attentional Control
		Accuracy	Attentional Control
		Median RT	Attentional Control
		RT Variability	Attentional Control
		Impulsivity Score	Inhibition
Augustinova et al. (2022)	Stroop Colour Word Task	Stroop Interference Score (Incon vs Neutral) (RT)	Attentional Control
		Stroop Interference Score (Incon vs Neutral) (Error Rate)	Attentional Control
		Semantic Interference Score (Incon vs Neutral) (RT)	Attentional Control
		Semantic Interference Score (Incon vs Neutral) (Error Rate)	Attentional Control
Bailey et al. (2018)	Digit Span Backwards	Total Correct	Working Memory
	Stroop Colour Word Task	Stroop Interference Score (Incon vs Con) (RT)	Attentional Control
Benedetti et al. (2023)	Go/No-go Task	% Incorrect No-go trials	Inhibition
		Mean RT (go trials)	Vigilance/ Sustained Attention
		% Correct (go trials)	Vigilance/ Sustained Attention
Berman et al. (2008)	Attention Network Test	Alerting Network Score	Vigilance/ Sustained Attention
		Orienting Network Score	Orienting
		Executive Network Score	Attentional Control
	Digit Span Backwards	Total Correct	Working Memory
Berman et al. (2012)	Digit Span Backwards	Total Correct	Working Memory
Berto (2005)	Sustained Attention to Response Task	Total Correct withheld responses	Inhibition
		Total Commission Errors	Inhibition
		d-prime	Vigilance/ Sustained Attention
		Mean RT	Vigilance/ Sustained Attention
Blum et al. (2019)	Stroop Colour Word Task	Stroop Interference Score (Incon vs Con) (RT)	Attentional Control
		Incongruent RT	Attentional Control
		Congruent RT	Vigilance/ Sustained Attention
Bodin and Hartig (2003)	Digit Span (Forward & Backward)	Combined Total	Working Memory
	Symbol Digit Modalities Task	Total Correct	Processing Speed
Boere et al. (2023)	Visual Oddball Task	Mean RT	Vigilance/ Sustained Attention
		Mean Errors	Vigilance/ Sustained Attention
Bourrier et al. (2018)	Digit Span Backwards	Total Correct	Working Memory
	Raven's Progressive Matrices	Total Correct	Logic/Critical Thinking
Bratman et al. (2015)	Digit Span Backwards	Total Correct	Working Memory
	Operation Span Task	Total Correct	Working Memory
	Change Detection Task	Capacity	Working Memory
	Attention Network Test	Executive Network Score	Attentional Control
Cassarino et al. (2019a)	Sustained Attention to Response Task	Total Accuracy	Inhibition
		Lures (No-go) Accuracy	Inhibition
		d-prime	Vigilance/ Sustained Attention
		Mean RT	Vigilance/ Sustained Attention
Cassarino et al. (2019b)	Sustained Attention to Response Task	Total Commission Errors	Inhibition
		d-prime	Vigilance/ Sustained Attention
		Mean RT	Vigilance/ Sustained Attention
Chen et al. (2020)	Necker Cube Pattern Control Task	Total Reversals	Attentional Control
Chow and Lau (2015)	Graduate Record Exam	Total Correct	Logic/Critical Thinking
Collado and Manrique (2020)	Digit Span (Forward & Backward)	Combined Total	Working Memory
Craig et al. (2015)	Sustained Attention to Response Task	Lures/No-go Accuracy	Inhibition
Crossan and Salmoni (2021)	Necker Cube Pattern Control Task	Total Reversals	Attentional Control
	Digit Span Backwards	Total Incorrect	Working Memory
de Brito et al. (2019)	Digit Span Backwards	Total Correct	Working Memory
Delgado and Zamarron (2019)	Cancellation Test	RT	Visual Tracking
		Total Correct	Visual Tracking
		Omissions	Visual Tracking
Emfield and Neider (2014)	Attention Network Test	Executive Network Score	Attentional Control
	Functional Field of View Task	RT	Orienting
		Accuracy	Orienting
	Digit Span Backwards	Capacity	Working Memory

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Table 2 (*continued*)

Author (Year)	Task	Outcome	Cognitive Domain
Engell et al. (2020)	Deary-Liewald Choice RT Task	Mean RT	Vigilance/ Sustained Attention
Evensen et al. (2015)	Reading Span Task	Total Correct	Working Memory
Gamble et al. (2014)	Attention Network Test	Executive Network Score	Attentional Control
		Orienting Network Score	Orienting
		Alerting Network Score	Vigilance/ Sustained Attention
	Digit Span Backwards	Total Correct	Working Memory
Geniole et al. (2016)	Stroop Colour Word Task	Stroop Interference Score (Incon vs Con) (RT)	Attentional Control
Gidlow et al. (2016)	Digit Span Backwards	Capacity	Working Memory
Greenwood and Gatersleben (2016)	Necker Cube Pattern Control Task	Mean Total Reversals	Attentional Control
Han (2017)	Forward Spatial Span	Total Correct	Basic Attention Span
	Digit Span Backwards	Total Correct	Working Memory
Hartanto et al. (2023)	Operation Span Task	PCU Score	Working Memory
Hartig et al. (2003)	Necker Cube Pattern Control Task	Mean Total Reversals	Attentional Control
Hicks et al. (2020)	Sustained Attention to Response Task	Total Commission Errors	Inhibition
		Omission Errors	Vigilance/ Sustained Attention
		Mean RT	Vigilance/ Sustained Attention
Ibrahim et al. (2022)	Alternative Uses Task	Total Alternative uses identified	Creativity
	Similarities Task	Total Similarities	Creativity
Jenkin et al. (2018)	Stroop Colour Word Task	Total Correct	Attentional Control
	Delay of Gratification Task	Total Delays	Inhibition
	Symbol Digit Modalities Task	Total Correct	Processing Speed
Jiang et al. (2019)	Digit Span (Forward & Backward)	Combined Total	Working Memory
Jiang et al. (2023)	Sustained Attention to Response Task	Total Accuracy	Inhibition
		Mean RT	Vigilance/ Sustained Attention
Jin et al. (2022)	Digit Span Backwards	Total Correct	Working Memory
Johansson et al. (2011)	Symbol Substitution Test	Total Correct	Processing Speed
Johnson et al. (2019)	Combined Attention Systems Test	Endogenous Executive Network Score (RT)	Attentional Control
		Endogenous Executive Network Score (Accuracy)	Attentional Control
		Exogenous Executive Network Score (RT)	Attentional Control
		Exogenous Executive Network Score (Accuracy)	Attentional Control
		Endogenous Spatial Stroop Score (RT)	Attentional Control
		Endogenous Spatial Stroop Score (Accuracy)	Attentional Control
		Exogenous Spatial Stroop Score (RT)	Attentional Control
		Exogenous Spatial Stroop Score (Accuracy)	Attentional Control
		Endogenous Orienting Network Score (RT)	Orienting
		Endogenous Orienting Network Score (Accuracy)	Orienting
		Exogenous Orienting Network Score (RT)	Orienting
		Exogenous Orienting Network Score (Accuracy)	Orienting
		Endogenous Alerting Network Score (RT)	Vigilance/ Sustained Attention
		Endogenous Alerting Network Score (Accuracy)	Vigilance/ Sustained Attention
		Exogenous Alerting Network Score (RT)	Vigilance/ Sustained Attention
		Exogenous Alerting Network Score (Accuracy)	Vigilance/ Sustained Attention
Johnson et al. (2021)	Attention Network Test	Alerting Network Score	Vigilance/ Sustained Attention
		Orienting Network Score	Orienting
		Executive Network Score	Attentional Control
Johnson et al. (2022)	Sustained Attention to Response Task	Total Commission Errors	Inhibition
		Mu	Vigilance/ Sustained Attention
		Sigma	Vigilance/ Sustained Attention
		Tau	Vigilance/ Sustained Attention
		Omission Errors	Vigilance/ Sustained Attention
Jones et al. (2021)	Digit Span Backwards	Unspecified	Working Memory
Jung et al. (2017)	Stroop Colour Word Task	Incongruent RT	Attentional Control
		Incongruent Error Rate	Attentional Control
		Non-switched RT	Attentional Control
		Non-switched Error Rate	Attentional Control
		Switched RT	Cognitive Flexibility
		Switched Error Rate	Cognitive Flexibility
		Congruent RT	Cognitive Flexibility
		Congruent Error Rate	Cognitive Flexibility

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Table 2 (*continued*)

Author (Year)	Task	Outcome	Cognitive Domain
	Multi-Source Interference Task	Congruent RT	Vigilance/ Sustained Attention
		Congruent Error Rate	Vigilance/ Sustained Attention
		Incongruent RT	Attentional Control
		Incongruent Error Rate	Attentional Control
	Digit Span Forwards	Capacity	Basic Attention Span
	Trail Making Test A	RT	Visual Tracking
	Trail Making Test B	RT	Cognitive Flexibility
Kimura et al. (2021)	Sustained Attention to Response Task	Mean RT	Vigilance/ Sustained Attention
		Hit-Rate (Go trial accuracy)	Vigilance/ Sustained Attention
Kimura et al. (2023)	Alternative Uses Task	Fluency Score	Creativity
		Flexibility Score	Creativity
		Originality Score	Creativity
Laumann et al. (2003)	Attention Orienting Task	RT (Exogenous Valid)	Orienting
		RT (Exogenous Invalid)	Orienting
		RT (Endogenous Valid)	Orienting
		RT (Endogenous Invalid)	Orienting
Lee et al. (2015)	Sustained Attention to Response Task	Total Commission Errors	Inhibition
		Mean RT	Vigilance/ Sustained Attention
		SD of RT	Vigilance/ Sustained Attention
		Omission Errors	Vigilance/ Sustained Attention
Lee et al. (2022)	Trail Making Test B	RT	Cognitive Flexibility
Li et al. (2022)	Alternative Uses Task	Total Creativity Score	Creativity
	Test for Creative Thinking	Total	Creativity
	Compound Remote Associates Task	Total	Creativity
	Raven's Progressive Matrices	Total Correct	Logic/Critical Thinking
Lin et al. (2014)	Capacity	Digit Span Backwards	Working Memory
Lymeus et al. (2017)	Letter Digit Substitution Test	Total Correct	Processing Speed
Ma and Shu (2018)	Visual Search Task	Total Correct	Visual Tracking
Manchon and Simunic (2023)	Trail Making Test A	RT	Visual Tracking
	Trail Making Test B	RT	Cognitive Flexibility
	Digit Span Forwards	Unspecified	Basic Attention Span
Morton et al. (2017)	Digit Span Backwards	Total Correct	Working Memory
Neilson et al. (2021)	Sustained Attention to Response Task	Total Correct withheld responses	Inhibition
		d-prime	Vigilance/ Sustained Attention

Table 2 (*continued*)

Author (Year)	Task	Outcome	Cognitive Domain
		Mean RT	Vigilance/ Sustained Attention
Raanaas et al. (2011)	Reading Span Task	Total Correct	Working Memory
		Total Correct (Same order)	Working Memory
Roe et al. (2020)	Deary-Liewald Simple RT Task	Mean RT	Vigilance/ Sustained Attention
Rogerson and Barton (2015)	Digit Span Backwards	Unspecified	Working Memory
Rogerson et al. (2016)	Digit Span Backwards	Total Correct	Working Memory
Sahlin et al. (2016)	Necker Cube Pattern Control Task	Total Reversals	Attentional Control
Schertz et al. (2022)	Dual n-back Task	A-prime (2-back)	Working Memory
		A-prime (3-back)	Working Memory
Shin et al. (2011)	Trail Making Test B	RT	Cognitive Flexibility
Tanaka et al. (2013)	Advanced Trail Making Test	Total Trials (Speed)	Visual Tracking
		Errors (Accuracy)	Visual Tracking
Trammell and Aguilar (2021)	Digit Span Forwards	Product Score	Basic Attention Span
	Digit Span Backwards	Product Score	Working Memory
	Trail Making Test A	RT	Visual Tracking
	Trail Making Test B	RT	Cognitive Flexibility
	Auditory Verbal Learning Task	Recall	Working Memory
		d-prime	Working Memory
Triguero Mas et al. (2017)	Digit Span Backwards	Total Correct	Working Memory
Van Hedger et al. (2019)	Digit Span Backwards	Total Correct	Working Memory
	Dual n-back	d-prime	Working Memory
Wade et al. (2020)	Rapid Visual Information Processing	RT	Vigilance/ Sustained Attention
		Accuracy	Vigilance/ Sustained Attention
	Spatial Working Memory Test	Errors	Working Memory
		Strategy	Working Memory
Wang et al. (2016)	Digit Span Backwards	Unspecified	Working Memory
Wang et al. (2018)	Taylor Aggression Paradigm	Provoked Aggression Score	Inhibition
		Unprovoked Aggression Score	Inhibition
Wang et al. (2021)	Symbol Digit Modalities Task	Total Correct	Processing Speed
Zhang et al. (2017)	Complement Test	Total Pairs Identified	Visual Tracking
Zhang, G. et al. (2023)	Stroop Colour Word Task	Stroop Interference Score (Incon vs Con) (RT)	Attentional Control
		Stroop Interference Score (Incon vs Con) (Accuracy)	Attentional Control
Zhang, D. et al. (2023)	Perception/ Attention Function Battery	RT	Vigilance/ Sustained Attention

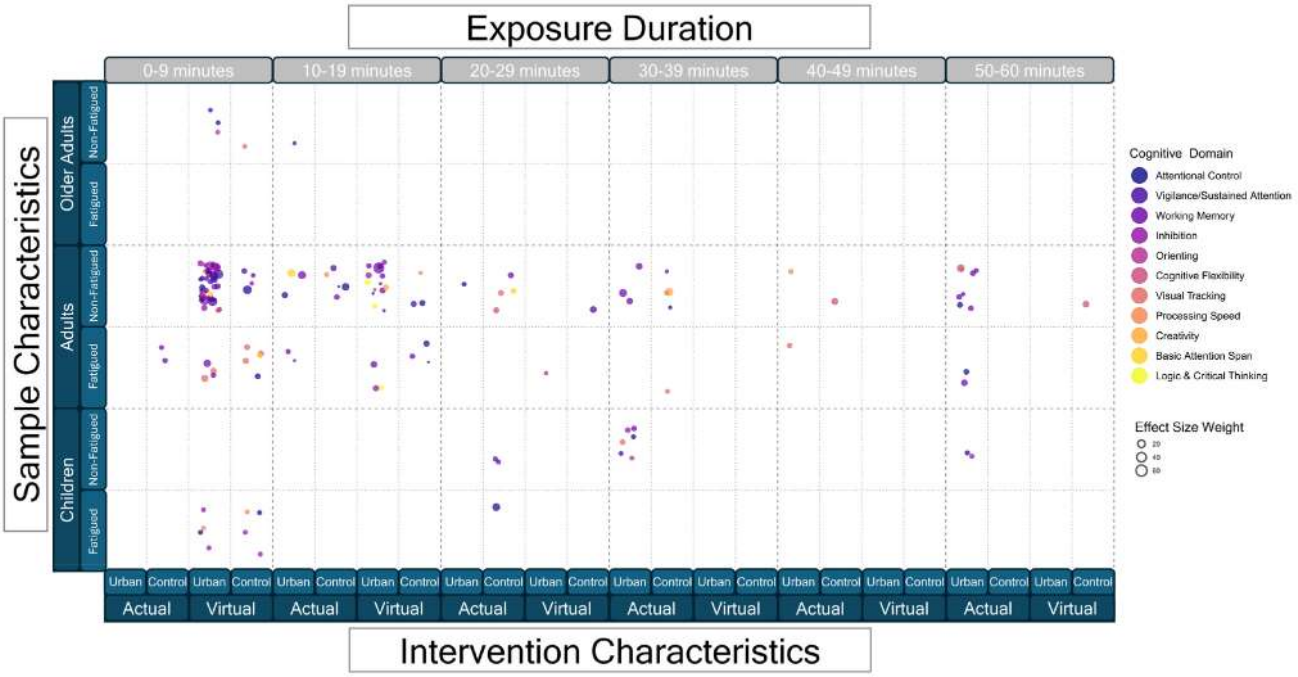


Fig. 3. Visualisation of extracted evidence across several key heterogeneity factors.

the double-count reduction process.

Most study interventions were sedentary (59%). Active-exposure studies included 11 with exercise, 17 with walking, and one design incorporating walking and exercise. Three studies did not specify an exposure activity but were unlikely to be sedentary (e.g., school recess free play). Nature exposures mostly consisted of full nature scenes (89%) through actual immersions or virtual presentation of the scene. Nine studies involved otherwise Urban/Built or control settings that had some degree of nature added to it; greenery added to urban streets or city rooftops (Jiang et al., 2023; Lee et al., 2015; Lin et al., 2014), pot plants in office spaces (Evensen et al., 2015; Ma & Shu, 2018; Raanaas et al., 2011), a house with relative openness to the nature outside (Tanaka et al., 2013), and nature sounds played in an indoor setting (Ma & Shu, 2018; Van Hedger et al., 2019). Comparator settings were mostly Urban/built scenes (66%) as opposed to control scenes. Overall, there were 86 comparator settings within the 80 studies, with 6 studies having nature groups contrasted against both urban and control settings (Emfield & Neider, 2014; Jenkin et al., 2018; Ma & Shu, 2018; Neilson et al., 2021; Rogerson & Barton, 2015).

In 20 studies, additional manipulations occurred alongside the environmental interventions. These included instructions regarding breathing or relaxation techniques (Blum et al., 2019; Sahlin et al., 2016), mindfulness practice (Lymeus et al., 2017) and imagery scripts (Chow & Lau, 2015), instructions to ruminate (Berman et al., 2012) or count to 1000 (Chen et al., 2020) during exposures, the presence of other friends, strangers, phones or laptops during exposures (Greenwood & Gatersleben, 2016; Jiang et al., 2019; Johansson et al., 2011; Rogerson et al., 2016; Wang et al., 2016), interactive or perturbated exposures (Crossan & Salmoni, 2021; Manchon & Simunic, 2023), awe vs mundane setting exposures (Collado & Manrique, 2020), intermediate cognitive assessments during exposures (Hartig et al., 2003), varied amounts or colour of nature (Lin et al., 2014; Zhang, Jin, et al., 2023), inanimate objects present in control exposures (Evensen et al., 2015) and the manipulation of salient social identity (Morton et al., 2017) or cognitive strategy (Bourrier et al., 2018). Some of these were applied to all participants, while others applied only to single conditions.

Regarding fatigue and cognitive demand, 58 studies administered a single cognitive test, while 22 studies administered two or more tests at pre/post-test. The order of these task batteries was sometimes counterbalanced, randomised, or matched. Experimental fatigue induction occurred in 23 studies (29%). Two of these were between-subjects manipulations (Hartig et al., 2003; Wang et al., 2018). Hartig et al. placed a long cognitive fatiguing phase after pre-testing. Although not all studies have cognitive testing *immediately* pre/post intervention, this method marked a qualitative deviation from other pre-to-post change scores in the analyses. As such, that specific contrast from the loading task arm of Hartig et al. (2003) was omitted from meta-analysis.

Acute exposure durations ranged from 40-s (Lee et al., 2015) to 60-min (e.g., Schertz et al., 2022). Triguero-Mas et al. (2017) included a longer exposure, but their design involved a lengthy measurement phase after a 30-min exposure, before an additional 3-h exposure. We opted to include the initial 30-min exposure to avoid risk of carryover effects and double inclusion of data. Overall, the distribution of exposure durations in experimental research has a slight skew towards shorter durations (Fig. 4). The mean exposure duration was 18-min, while the median was 10-min. These two values are marked for reference in all exposure duration regression figures by dashed vertical lines.

3.1.2. *Quality appraisal and risk of bias*

Generally, a lack of information regarding the randomisation process and pre-registered analysis plans generated concerns in most studies (see supplementary material S3). Non-computer-based task administration (e.g., experimenter delivered DSB) also represented some concern due to possible subconscious differences in cadence or pitch in task delivery when a participant's intervention condition is known. Overall, we downgraded most algorithm-generated risk of bias scores by one category based on the apparent experimental rigour and cognisance of publication-era norms.

The quality of all studies was rated as moderate/high (52.6%–97.4%; see supplementary material S2). We note that not all items reflect study quality in all contexts, particularly for studies not explicitly researching attention restoration. Indeed, given it is unclear that designs involving cognitive fatigue are of higher quality than those without, the 'demonstrated need for restoration' item may be better suited as a descriptive subgrouping variable than a quality indicator. In general, study quality scoring was lowered by a lack of reporting *a priori* power calculations and randomisation processes, and over representation of non-generalisable student samples.

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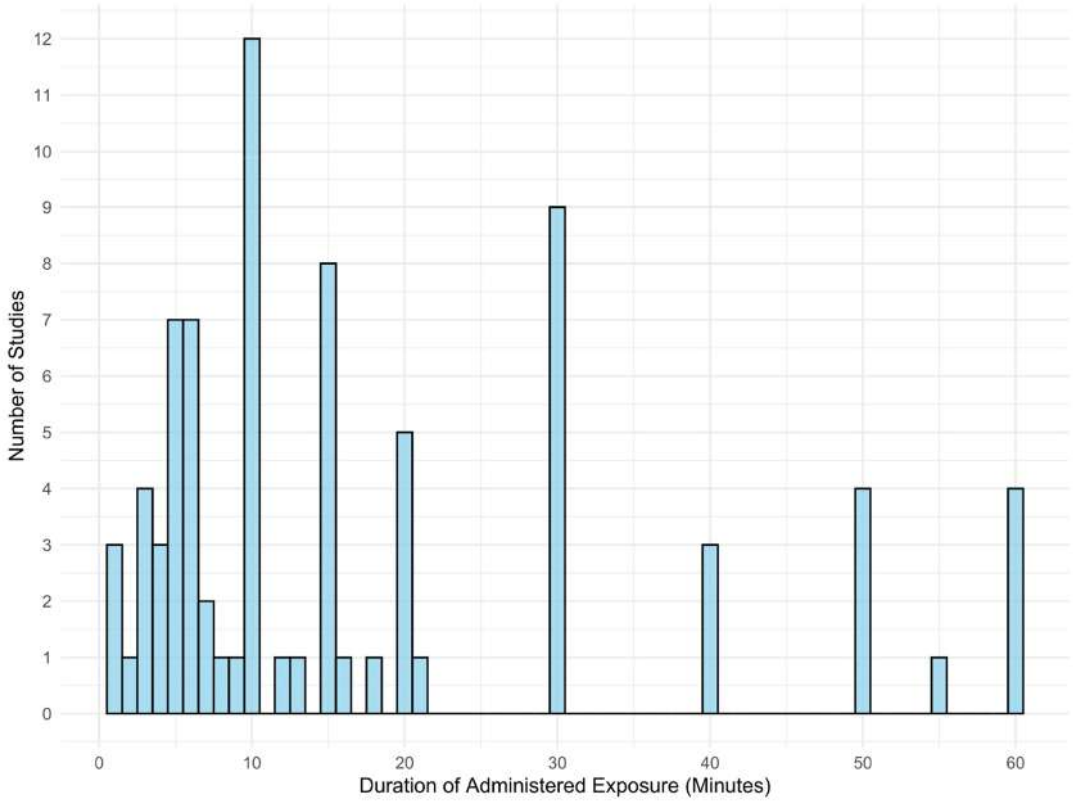


Fig. 4. Histogram visualising the number of studies administering different exposure durations.

3.2. Part 2: meta-analyses

After combination of groups to reduce double counting, a total of 273 natural vs non-natural contrasts were included in the meta-analyses. There were 39 unique cognitive tests administered across the 80 studies, yielding 123 unique cognitive outcomes. These 123 outcomes were categorised to represent 11 broad cognitive domains (see section 2.5).

3.2.1. Natural vs non-natural effects within cognitive domains

The most observations are seen for domains of Attentional Control (61 natural vs non-natural contrasts), Vigilance/Sustained Attention (52 contrasts) and Working Memory (51 contrasts). Less researched are Inhibition (30), Orienting (24), Cognitive Flexibility (15) and Visual Tracking (14). Domains that have received little attention are Processing Speed (9) Creativity (8), Basic Attention Span (5) and Logic/Critical Thinking (4).

3.2.1.1. Publication bias within cognitive domains. Standard and contour-enhanced funnel plots for each domain are provided in supplementary material S9. Small-sample asymmetry typical of publication bias was visually observed in the Creativity and Inhibition domains, and to a lesser degree in Visual Tracking. Egger's Tests (Table 3) did not statistically confirm asymmetry, although these analyses within domains may have limited power. As such, the distinct asymmetry generated by two large statistical outliers among Inhibition effects ($g = 2.36$; $g = 3.13$), and two outlying Creativity effects ($g = 2.06$; $g = 2.24$) were removed for the reported meta-analyses. These effects are also visually outlying when viewing the funnel plot of all 273 cognitive outcomes together (Fig. 8). Within the Creativity domain, removal of these effects resulted in model non-convergence, thus no reliable effect estimates were attainable for creativity. Most selective-reporting-corrected estimates were marginally smaller than the focal CHE model effect sizes, and generally do not suggest that inclusion of unknown 'file-drawered' research would lead to drastic differences in magnitude or interpretation (Supplementary Material S7).

Table 3
Egger's Test Output for the Overall Model and each Within-Domain Model.

	Egger's Regression Tests			
Cognitive Domain	Intercept	p-value	Slope	p-value
Attentional Control	−0.06	0.820	0.82	0.390
Basic Attention Span	0.18	0.795	−0.63	0.820
Cognitive Flexibility	1.40	0.266	−5.28	0.221
Creativity	−3.55	0.539	18.41	0.474
Creativity[a]	0.09	0.940	−0.01	0.999
Inhibition	0.20	0.745	0.29	0.883
Inhibition[a]	0.44	0.305	−1.04	0.458
Logic/Critical Thinking	1.60	0.576	−4.92	0.621
Orienting	0.04	0.938	0.21	0.885
Processing Speed	−1.66	0.353	4.89	0.302
Vigilance/Sustained Attention	0.00	0.993	0.38	0.633
Visual Tracking	−0.76	0.237	3.30	0.135
Working Memory	0.16	0.488	0.29	0.719
All Cognitive Outcomes	0.05	0.716	0.59	0.212
All Cognitive Outcomes[a]	0.08	0.531	0.36	0.416

Note.
[a] Outlier-removed results.

3.2.1.2. Sensitivity analyses within cognitive domains. Variance component (σ^2) estimates displayed some sensitivity to changing constant sampling correlation values. Within-study variance typically increased with increasing sampling correlations, while between study variance decreased (supplementary material S6b). There were typically slight increases in heterogeneity at higher values of assumed constant sampling correlation, however, mean SMDs remained extremely stable (supplementary material S6a). Several of these domains included <10 observations, making heterogeneity estimates unreliable (Borenstein,

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2019).

3.2.1.3. Pooled SMDs within cognitive domains. Table 4 contains meta-analytic output for all models. Forest plots displaying all outcomes generating each pooled effect are provided in supplementary material S10. The mean SMD in all domains favoured nature conditions, ranging from trivial to small effect sizes. Creativity outcomes reflected a large, but unreliable SMD favouring nature ($g = 0.75$) driven by two large outlier effects from single study. As such, Creativity results are not interpreted further.

3.2.1.3.1. Attentional control. Across 19 studies, 61 contrasts were observed from six cognitive tests, encompassing 29 unique outcomes (Supplementary Material S4). The mean SMD was very small but indicates the true mean effect reliably favours nature conditions. All heterogeneity attributable to true effect variance was occurring within-studies rather than between, while overall there was considerable heterogeneity, with 95 % PIs indicating the true effect could plausibly fall ±0.66 standard deviations of the observed mean SMD (Table 4).

3.2.1.3.2. Vigilance/sustained attention. Across 23 studies, 52 contrasts were observed from 10 cognitive tests, encompassing 22 unique outcomes (S4). The mean SMD was very small and the true mean effect can plausibly favour either natural or non-natural comparator conditions. All heterogeneity attributable to true effect variance was occurring within-studies, and 95 %PIs reflect substantial heterogeneity (Table 4).

3.2.1.3.3. Working memory. Across 33 studies, 51 contrasts were observed from eight cognitive tests, encompassing 18 unique outcomes (S4). The mean SMD was small, but indicates the true mean effect reliably favours nature conditions. All heterogeneity attributable to true effect variance was occurring within-studies, and 95 %PIs reflect substantial heterogeneity (Table 4).

3.2.1.3.4. Inhibition. After removal of two outliers, 28 contrasts in 16 studies were observed from five cognitive tests, encompassing 10 unique outcomes (S4). The mean SMD was very small and the true mean effect can plausibly favour either natural or non-natural comparator conditions. All heterogeneity attributable to true effect variance was occurring within-studies, and 95 %PIs reflect substantial heterogeneity (Table 4).

3.2.1.3.5. Orienting. Across 6 studies, 24 contrasts were observed from four cognitive tests, encompassing 15 unique outcomes (S4). The mean SMD was trivial and the true mean effect can plausibly favour either natural or non-natural comparator conditions. All heterogeneity attributable to true effect variance was occurring within-studies, and 95 %PIs reflect substantial heterogeneity (Table 4).

3.2.1.3.6. Cognitive Flexibility. Across five studies, 15 contrasts were observed from two cognitive tests, encompassing five unique outcomes (S4). The mean SMD was very small and the true mean effect can plausibly favour either natural or non-natural comparator conditions. Most heterogeneity was occurring between studies, and 95 %PIs reflect substantial heterogeneity (Table 4).

3.2.1.3.7. Visual Tracking. Across nine studies, 14 contrasts were observed from six cognitive tests, encompassing eight unique outcomes (S4). The mean SMD was very small and the true mean effect can plausibly favour either natural or non-natural comparator conditions. All estimated variance in this model is attributable to sampling variance, with minimal overall heterogeneity reflected by the 95 %PIs (Table 4).

3.2.1.3.8. Processing Speed. Across five studies, nine contrasts were observed stemming from a single unique outcome, from what we deem to be a single cognitive test, despite appearing in the literature under several different names (S4). The mean SMD was trivial and the true mean effect can plausibly favour either natural or non-natural comparator conditions. Most heterogeneity attributable to true effect variance was occurring within-studies, and 95 %PIs reflect substantial heterogeneity (Table 4).

3.2.1.3.9. Basic Attention Span. Across four studies, five contrasts were observed from two cognitive tests, encompassing four unique outcomes (S4). The mean SMD was trivial and the true mean effect can plausibly favour either natural or non-natural comparator conditions. All estimated variance in this model is attributable to sampling variance, as such, there was less overall heterogeneity, reflected by tighter 95 % PIs (Table 4).

3.2.1.3.10. Logic/Critical Thinking. Across three studies, four contrasts were observed from two cognitive tests, encompassing two unique outcomes (S4). The mean SMD was small and the true mean effect can plausibly favour either natural or non-natural comparator conditions. All heterogeneity attributable to true effect variance was occurring within-studies, and 95 %PIs reflect moderate heterogeneity (Table 4).

3.2.1.4. Exposure duration within cognitive domains. Of the 11 domains, only seven contained the requisite observations allowing for moderator analysis using meta-regression. Orienting, Vigilance/Sustained Attention and Visual Tracking domains saw better model fit from linear meta-regressions relative to quadratic, although these domains tended to lack observations for longer durations which precludes the hypothesised quadratic effect (Fig. 5). A quadratic meta-regression model provided better fit than a linear model for Attentional Control, Cognitive Flexibility, Inhibition, and Working Memory (Fig. 6). Model fit was typically not improved by exposure duration meta-regressions relative to the unmoderated model, although differences were mostly negligible (Table 5). The only statistically significant associations between exposure duration and SMD were the quadratic relationships modelled in the Cognitive Flexibility and Working Memory domains.

The shape of the significant quadratic relationship for Working Memory, and the non-significant Attentional Control and Inhibition quadratic models, were consistent with H_2. Mean SMDs increase in positivity (favouring natural settings) with earlier exposure durations, before peaking and reducing in size as exposure durations increase (Fig. 6a, b, c). Conversely, the significant Cognitive Flexibility quadratic

Table 4
CHE-RVE Output from Independent Meta-Analyses Conducted for each Cognitive Domain.

		Between-group (Natural vs Non-Natural) estimates				Heterogeneity		
Cognitive Domain	*n* (k)	hedges *g* [95 %CI]	*p*	95 %PI	PI ±	I^2_{level2}	I^2_{level3}	I^2_{total}
Attentional Control	61 (19)	0.16 [0.02, 0.30]	0.028	[-0.51, 0.83]	0.67	57.51	0	57.51
Vigilance/Sustained Attention	52 (23)	0.10 [-0.05, 0.25]	0.164	[-0.51, 0.72]	0.62	56.53	0	56.53
Working Memory	51 (33)	0.24 [0.13, 0.35]	**<.001**	[-0.29, 0.77]	0.53	47.54	0	47.54
Inhibition	28 (16)	0.14 [-0.02, 0.31]	0.087	[-0.75, 1.04]	0.90	71.00	0	71.00
Orienting	24 (6)	0.10 [-0.16, 0.35]	0.311	[-0.63, 0.83]	0.73	62.51	0	62.51
Cognitive Flexibility	15 (5)	0.10 [-0.48, 0.68]	0.653	[-0.80, 1.00]	0.90	9.27	58.86	68.13
Visual Tracking	14 (9)	0.17 [-0.08, 0.41]	0.153	[-0.23, 0.56]	0.39	25.47	0	25.47
Processing Speed	9 (5)	0.01 [-0.56, 0.58]	0.963	[-0.79, 0.81]	0.8	35	15.75	50.75
Basic Attention Span	5 (4)	0.03 [-0.26, 0.31]	0.779	[-0.21, 0.26]	0.23	0	0	0
Logic/Critical Thinking	4 (3)	0.23 [-0.47, 0.93]	0.300	[-0.23, 0.68]	0.45	21.91	0	21.91
Creativity[E]	8 (3)	0.75 [-2.16, 3.67]	0.382	[-1.87, 3.37]	2.62	0	95.95	95.95

Note. [E] = unreliable model excluded from interpretation, **bold** = statistically significant to multiplicity adjusted alpha of 0.01.

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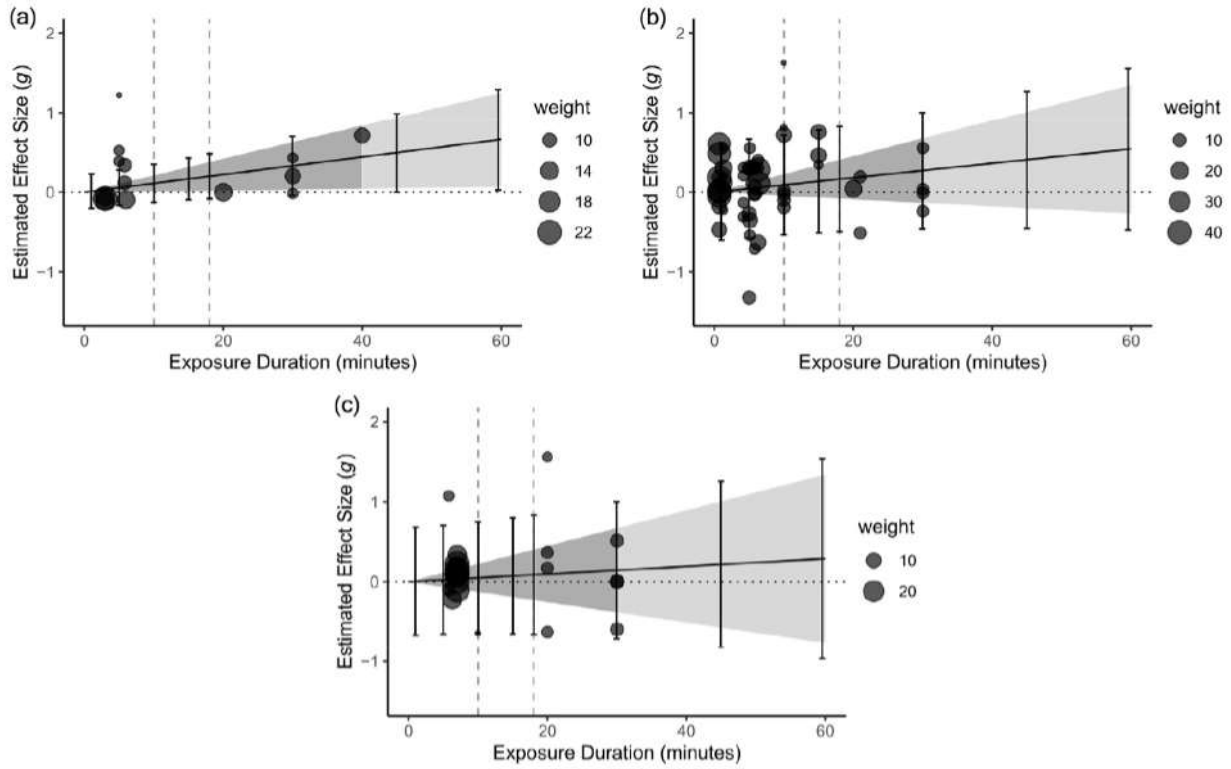


Fig. 5. Linear relationship between exposure duration and mean SMD in the domains of visual tracking (a), vigilance/sustained attention (b) and orienting (c). *Note.* Line = predicted mean SMD, Points = observed weighted SMDs, dark shading = 95 %CI estimates, light shading = 95 %CI estimates extrapolated beyond observed data range, Error Bars = 95 % prediction interval estimates, dashed vertical lines represent the median (10) and mean (18) exposure durations.

model saw opposite effect. Here, estimated mean SMDs predict negative values further decreasing (favouring non-natural settings) for shorter duration exposures, before peaking and favouring nature at longer exposure durations (Fig. 6d).

3.2.1.5. Subgroup analyses within cognitive domains. A total of 62 subgroup models were fitted across the 10 potential sources of methodological heterogeneity in the seven eligible cognitive domains. Informal observations of patterns across domains are described below and all statistics are presented in supplementary material S11.

For the subgroup analyses of Type of Exposure, Immersion of Exposure, Density of Nature Exposure, Exercise, Comparator Setting, Age, Cognitive Tests, there were no reliable differences in between-subgroup mean SMD (i.e., no evidence for differences in restoration that were dependent on subgroup membership). In contrast, the subgroup variables of Clinical Samples (clinical vs non-clinical), Other Manipulation (yes vs no) and Fatigue Induction (fatigued vs non-fatigued) each saw reliable between-subgroup differences in mean SMD in at least one cognitive domain.

Three domains had requisite observations to run 'Other Manipulation' subgroup analyses. Only Attentional Control displayed a reliable difference in mean SMD, whereby SMDs from contrasts occurring with additional manipulations were, on average, greater than those without ($\Delta g = 0.31$ [0.08, 0.55], $p = .016$). By including this subgroup variable in the model, it improved model fit relative to the unmoderated Attentional Control meta-analysis.

Five domains had requisite observations to run 'Clinical Samples' subgroup analyses. Only Cognitive Flexibility displayed a reliable difference in mean SMD, such that SMDs were on average higher for non-clinical samples ($\Delta g = 0.22$ [0.16, 0.29], $p = .008$), but inclusion of this variable did not improve model fit relative to the unmoderated analysis.

Five domains had requisite observations to compare fatigued and non-fatigued subgroups. Generally, the subgroup *n* were substantially greater for non-fatigued contrasts. Two domains displayed a reliable difference. In the Inhibition ($\Delta g = 0.46$ [0.21, 0.72], $p = .011$) and Working Memory domains ($\Delta g = 0.26$ [0.04, 0.47], $p = .024$), fatigued participants' mean SMD reliably favoured nature conditions more than for non-fatigued participants. In both cases, including Fatigue Induction in the model provided a better model fit relative to the unmoderated meta-analyses.

3.2.2. Difference of environmental effects among cognitive domains

The SCE-RVE meta-analysis did not reveal statistically significant differences in SMD among the domains ($QM(df = 10) = 6.47$, $p = .775$). Two factors were speculated to contribute to this finding: high heterogeneity of effects 3.2.1 and disparity in observations among subgroups. Two additional exploratory analyses were thus conducted.

The CHE and SCE models were re-fitted with all statistical outliers removed from the dataset (29 effects; 13 above the pooled SMD and 16 below) which reduced heterogeneity by almost half (supplementary material S12). This reduction in heterogeneity did not substantially influence the SCE model results, ($QM(df = 10) = 8.42$, $p = .588$). Next, minimally observed cognitive domains were removed to leave a contrast of only the more balanced domains/subgroups, which again did not substantially influence the findings for test of subgroup differences (Supplementary Material S12b).

3.2.3. Natural vs non-natural environment effects across cognitive domains

Following the non-significant SCE analysis providing no *statistical* justification to distinguish effects between cognitive domains, the CHE analysis plan was also enacted on all cognitive outcomes together. Overall, effect sizes reliably favour nature conditions to a small degree ($g = 0.17$ [0.11, 0.23], $p < .001$). I^2 values indicate that the proportion of heterogeneity attributable to variance in true effects was 57.05 %, occurring wholly at level 2 (within studies). PIs around the pooled SMD (−0.45, 0.80) depict substantial heterogeneity, indicating that the true

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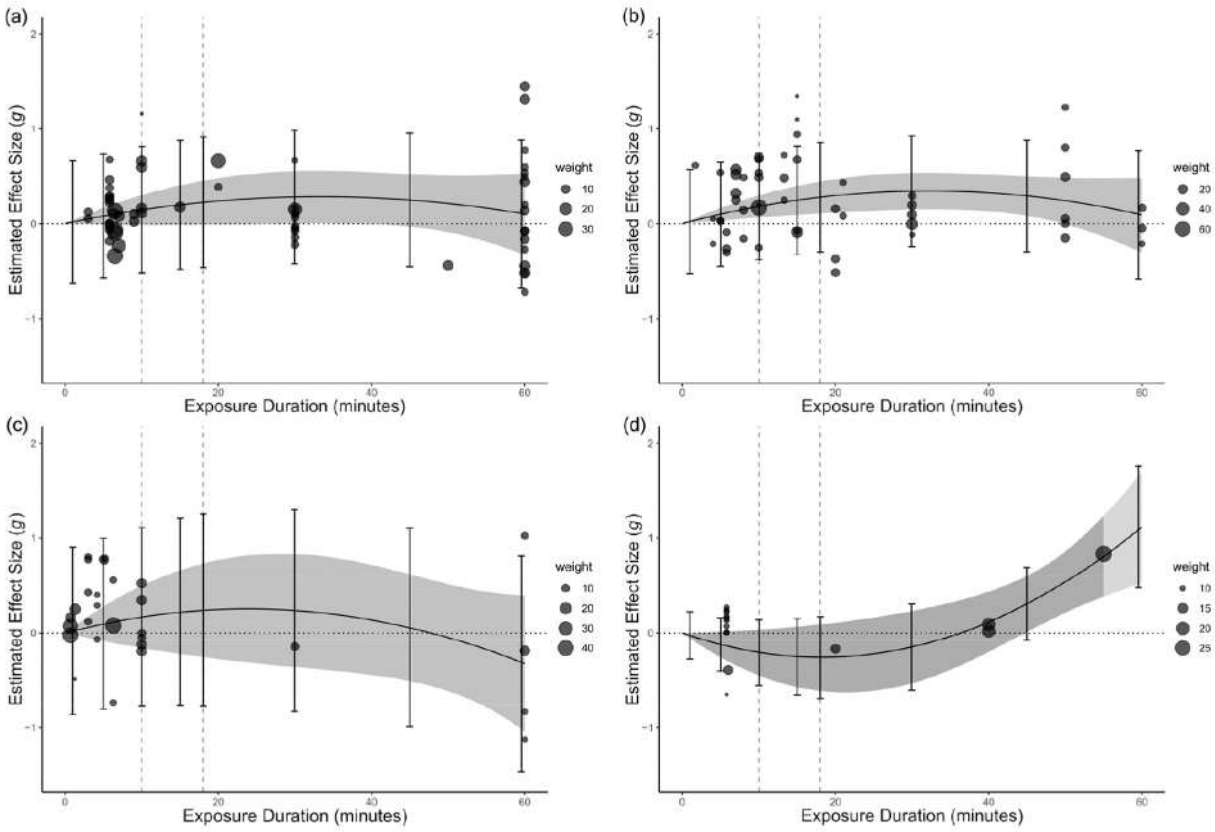


Fig. 6. Quadratic relationship between exposure duration and mean SMD in the domains of attentional control (a), working memory (b), inhibition (c) and cognitive flexibility (d).
Note. Line = predicted mean SMD, Points = observed weighted SMDs, dark shading = 95 %CI estimates, light shading = 95 %CI estimates extrapolated beyond observed data range, Error Bars = 95 % prediction interval estimates, dashed vertical lines represent the median (10) and mean (18) exposure durations.

Table 5
Model fit output for all unmoderated, linear and quadratic models in domains with requisite observations for meta-regression.

	Akaike Information Criterion		
	Unmoderated	Linear	Quadratic
All Cognitive Outcomes*	**297.22**	303.87▲	299.34▲
Cognitive Domain			
Attentional Control	**71.63**	72.88	72.11
Cognitive Flexibility	13.59	12.38	**09.73**▲
Inhibition	**47.61**	48.62	48.29
Orienting	**34.85**	35.00	35.64
Vigilance/Sustained Attention	**64.84**	64.94	65.26
Visual Tracking	15.37	**013.55**	15.42
Working Memory*	**53.69**	60.60	57.90▲

Note. AIC values only to be compared within rows, no other comparison is valid. **Bold** values indicate the best model fit of the three models. * = significant natural/non-natural difference. ▲ = significant moderating effect of exposure duration.

effect could plausibly fall ±0.63 standard deviations of the observed mean SMD. A forest plot including all outcomes is in supplementary material S13, and sensitivity analyses in supplementary material S6.

3.2.3.1. Exposure duration across cognitive domains. A quadratic meta-regression model was a better fit to the data than the linear model (Table 5). Depicted in Fig. 7, the overall quadratic effect of exposure duration on SMDs was statistically significant (QM($df = 2$) = 20.41, $p <$ .001) reflecting an increased favouring of nature with each minute of exposure duration (β = 0.017 [0.009, 0.024], $p <$ .001), which peaks and then decreases as exposure durations increase (β = −0.000 [−0.000, −0.000], p = .006). Model fit was best for the unmoderated meta-analysis despite the significant moderating effect (Table 5), although difference in AIC values was negligible (Cavanaugh & Neath, 2019). Further testing of this relationship in different population subgroups (see Supplementary Material S14 for model output and figures) generally mirrored the pattern of results described above across ages (<18s, Adults) and comparator settings (Urban/Control). A linear model was a better fit for Older Adults and Clinical Samples, though both of these subgroups tend to be under-observed, with primarily short-duration exposures (Fig. 3).

3.2.3.2. Subgroup moderators. Supplementary material S15 contains results from subgroup analyses when all cognitive outcomes are combined. There were no significant differences between mean SMDs in each subgroup contrast. One subgroup contrast reliably operated in one direction (marginal statistical significance after multiplicity correction), whereby fatigued participants had a higher SMD than non-fatigued participants (Δg = 0.19 [0.04, 0.33], p = .014). Further, this fatigue subgroup meta-regression was the best fit of the data, and an improvement on the initial unmoderated analysis.

3.2.3.3. Publication bias, sensitivity analyses & imputed change-variability. The funnel plot depicts a largely symmetrical distribution of effects with many points lying outside the pseudo-confidence intervals in both directions (Fig. 8). The within-domains outliers remain visually distinct, and are removed from this overall analysis to retain uniformity with domain-specific analyses. Supplementary material S6 contains sensitivity analyses plots, where patterns were similar to those observed within domains.

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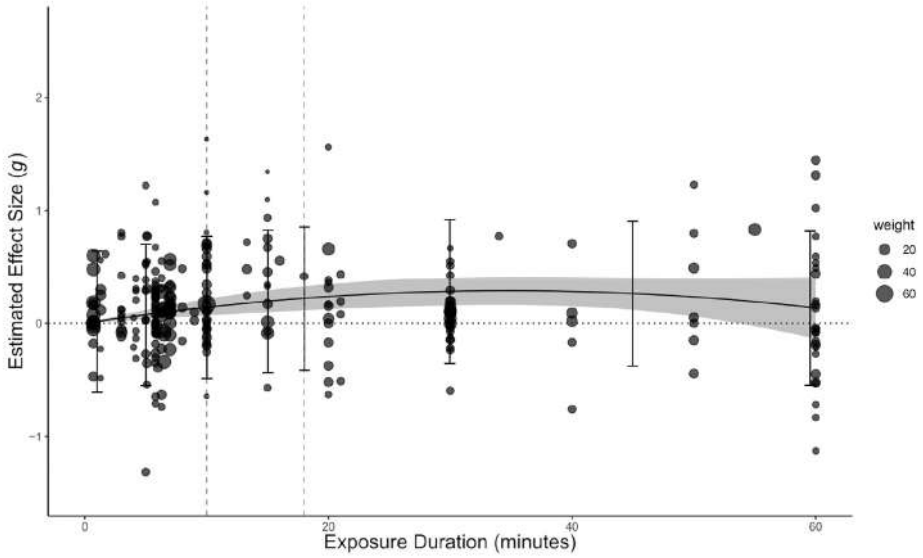


Fig. 7. Non-linear relationship between exposure duration and mean SMD

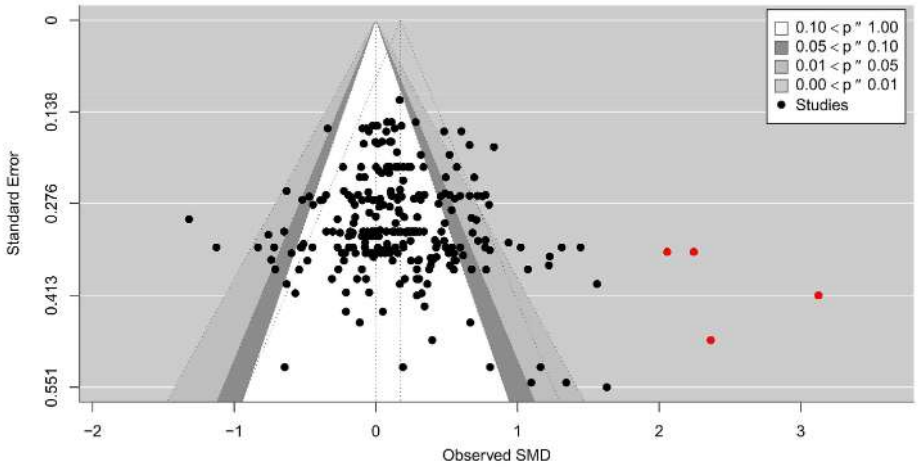


Fig. 8. Contour-enhanced funnel plot of all cognitive outcomes.
Note. Red dots indicate outliers identified within domains, thus also omitted from analysis of all outcomes combined.

According to Cochrane recommendations, the overall percentage of imputed change-SDs was high, though still represented the minority of observations (44 %) and studies (28 %). Subgroup analysis of all contrasts indicated no significant difference between SMDs generated using true change-variability vs using imputed change-variability statistics, despite imputed SMDs having a slightly lower pooled effect estimate ($\Delta g = 0.08$ [−0.04, 0.20], $p = .170$). As such, there is no evidence of the imputation method biasing results.

4. Discussion

This review synthesised the current scientific evidence from randomised, pre-post intervention experiments that detail the difference in restorative effects between natural and non-natural environments, and the influence of exposure durations on these effects. All *average* effect sizes favoured nature conditions, though only reliably for Working Memory and Attentional Control. Despite some variation in the pooled effect size magnitude across cognitive domains, these differences were not statistically significant. A consequent combining of outcomes from all cognitive domains together saw the effect significantly favour nature. Exposure duration moderated this effect size in a non-linear relationship, which also varied within specific domains.

4.1. Key findings

4.1.1. Observable, but non-significant, effect size magnitude differences between cognitive domains

The pooled effect estimates for all 11 cognitive domains favoured nature conditions, with sizes ranging from negligible to small (omitting Creativity). Working Memory had the largest effect, followed by Logic/Critical Thinking and Attentional Control. In contrast, effects were negligible for Processing Speed, Basic Attention Span, Vigilance/Sustained Attention, Orienting, Cognitive Flexibility, Visual tracking, and Inhibition. Regardless of magnitude, the four most observed domains (Attentional Control, Vigilance/Sustained Attention, Working Memory and Inhibition) predictably emerged with the most reliable effect estimates, with confidence intervals less than 0.16 standard deviations from their pooled effect. Of these, only the Working Memory and Attentional Control effects showed reliable favouring of nature conditions, with natural settings being significantly more restorative than non-natural comparators for Working Memory.

Consistent with H_1, informally comparing domain effect sizes tended to depict larger effects favouring nature in domains that encompass higher order cognitive functions, while the weaker effects occurred in domains that are less demanding or associated with bottom-up

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processing. This aligns with ART and observations from previous meta-analysis (Stevenson et al., 2018), however, informal comparisons of effect size magnitude have limited capacity to formally evidence the presence/absence of effects in different cognitive domains (Borenstein et al., 2021).

A formal statistical contrast of all cognitive domain effects contradicted H_1, whereby no effect sizes significantly differed from one another. Despite Working Memory having the largest (and only statistically significant) effect, this difference in effect size from the other cognitive domains was not statistically meaningful. Crucially, this suggests that quantitative differences of this size could occur even when no true qualitative difference exists between how nature exposures influence restoration in these domains. We thus cannot conclude with certainty that nature restores working memory any differently than it does any other cognitive domain – a more conservative interpretation relative to using only within-domain statistical significance to evidence the presence/absence of effects across domains (e.g., Stevenson et al., 2018).

However, an additional six years of ART literature have now been synthesised to produce a highly aligned finding to Stevenson et al. (2018). In particular, the contrast of two heavily researched cognitive domains in the ART literature, Working Memory and Vigilance/Sustained Attention, which are also generally representative of the top-down/bottom-up distinction of direction attention reliance and need for restoration (Ohly et al., 2016). As with Stevenson et al., we observed a significant Working Memory effect and no such effect for Vigilance/Sustained Attention. This is a quasi-replication not to be ignored. We can be confident the within-domain differences in our analysis are not merely attributable to the number of observations in each domain and resulting statistical power, as the Working Memory and Vigilance/Sustained Attention domains were virtually equally represented. Taken together (along with other Working Memory meta-analysis findings; Stenfors et al., 2019), we tentatively suggest that a nature restoration effect does exist and is larger for working memory, in comparison to vigilance/sustained attention outcomes where the presence of an effect is less clear. However, we stress again that this conclusion is descriptive only, not based on inferential statistics generated by the SCE analysis. It appears the effect sizes themselves, and thus the differences between them, were too small to statistically distinguish in our formal comparison.

4.1.2. Relationship between effect size and exposure duration

When all cognitive outcomes were analysed together, exposure duration significantly moderated effect size. Consistent with H_2, this effect was non-linear, such that there was an increasing nature benefit associated with increasing exposure durations – up to a point. Specifically, it appears the average nature benefit peaks around 30 min of exposure, before benefits may begin to subside – although effect estimates are less reliable at longer durations.

Within-domains, this relationship is seen again for Working Memory, Attentional Control and Inhibition, although of these, only Working Memory effects were significantly moderated by exposure duration. Cognitive Flexibility effects were also moderated by exposure duration, but in the opposite direction to H_2. We do not expect that Cognitive Flexibility has a truly opposite relationship with exposure duration compared to other domains and cognitive outcomes overall, but may be influenced by a highly weighted study with a large effect size favouring nature after a long (1-h) exposure. This highlights the observational nature of meta-regression and potential for spurious relationships to emerge which are unrelated to the moderator in question.

Several within-domain analyses predicted a linear relationship with exposure duration to be a better fit to the data than non-linear (Visual Tracking, Orienting and Sustained Attention/Vigilance). This contradicts H_2, and the restorative logic of ART more broadly. However, none of these linear effects were highly reliable estimates, and none contained effect sizes from exposures above 30–40 min. As such, these domains may be reflecting only the first half of the non-linear relationship described above. Overall, a non-linear relationship between exposure duration and effect size is more supported than a linear relationship.

4.1.3. Subgroup analyses & heterogeneity of effects

Only fatigue induction emerged from subgroup analyses as a factor influencing effect size magnitude with any consistency. Including fatigue induction produced the best model fit to the overall data, where the nature restoration effect was larger for fatigued samples. Nature benefits were also larger for fatigued samples within each domain, though these differences were only reliably greater for Inhibition and Working Memory. These findings align with theorised effects where domains with greater recruitment of directed attention will be more susceptible to fatigue and thus nature restoration. Unfortunately, there were no fatigued-samples in the Vigilance/Sustained Attention domain to allow for a contrast in more bottom-up cognitive domain. This reflects a surprising lack of data regarding a key premise of ART (cognitive fatigue) in one of the more highly researched ART outcomes.

The absence of other reliable subgroup effects was surprising, as these variables were expected to explain some degree of heterogeneity of effects – which was substantial across and within domains. Although *average* effects reliably favoured natural environments overall, as well as for Working Memory and Attentional Control domains specifically, the heterogeneity indicated a wide dispersion of effects, such that the true effect could plausibly favour either natural or non-natural settings. This remained so when exposure duration significantly moderated effect size. Exploratory analyses removing outliers reduced heterogeneity considerably, however, there was no basis for removal of these data beyond statistical non-conformity. Instead, these are likely valid experimental observations of what appears to be a highly heterogenous effect.

Interestingly, the prevailing tendency was for true effect variance to be occurring within-studies. This may explain the limited evidence of heterogeneity when comparing our subgroups, as they are primarily between-study manipulations. Aside from sampling variance that makes up approximately one third of overall heterogeneity, it appears differences within individual studies may be most pertinent to effect heterogeneity. Experiments regularly include several different cognitive tasks/outcomes that would fall in the same meta-analysis (e.g., Bratman et al., 2015; Johnson et al., 2022). This could lead to differences in intervention effects across metrics (e.g., raw totals, difference scores, response times, accuracy percentages and error rates), particularly where scores themselves can be correlated (e.g., speed/accuracy trade-offs; Bogacz et al., 2010).

Ohly et al. (2016) analysed effects on specific outcomes and observed non-significant heterogeneity, although, with so few observations available once analysing individual outcome measures, heterogeneity estimates are far less reliable (Borenstein, 2019, 2024). Future meta-analyses with even more data may opt to increase outcome specificity to quantify how dependent variable differences impact heterogeneity of nature restoration effects.

4.2. Heterogeneity in the ART context

A broader issue is how the substantial heterogeneity of effects matters to the interpretation of our results. Generally, meta-analysing a collection of effects is justified when they are 'sufficiently homogenous'; by synthesising evidence on apples and oranges one can void making a meaningful finding for either (Higgins et al., 2023). However, this analogy is only true if you believe the effects of each fruit (population) are of interest independently, otherwise, this combination can provide meaningful information about fruit in general (Higgins et al., 2023).

Some reviews, such as those on drug efficacy, must explore substantial heterogeneity because it indicates the drug may be harming some populations (if the PIs cross zero), they need to find out who these populations are and ensure they are not recommended the drug. In our context, outcomes are strictly positive (restoration), with values below

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zero merely indicating the benefit may be larger for a non-natural exposure in some populations. While we strive to fully explain effect heterogeneity in the same way, 'nature' is an extremely heterogeneous intervention, and heterogeneity of its effects may be an unsurprising byproduct. It is also not certain that determining the specific combination of niche individual/intervention factors that characterise the heterogeneity would translate to improved outcomes for the general population.

Our pooled effects can be highly generalisable and informative by depicting the expected *average* benefit of nature exposures, even with the caveat that effects could vary broadly from person-to-person in the real world. We currently have minimal evidence to explain this heterogeneity, after considerable exploration of straightforward population subgroup differences (e.g., adult vs children, actual vs virtual exposures, exercise vs sedentary, etc.). It is therefore plausible that heterogeneity is instead driven by a nuanced aggregate of these subgroup combinations, which may be further obscured by individual difference variables and variation in measurement techniques. In sum, dissecting the 'apples from oranges' in search of effect homogeneity may negatively impact the generalisability of research findings, where average effects of nature are quite reliable and have broader utility.

4.3. Strengths, limitations and future research

Strengths of this review include screening for high quality, randomised pre/post experimental outcomes from objective measures; analyses that appropriately handle hierarchical data structures and correlation of effects while maximising data inclusion; an expanded range of moderator analyses; performance change-statistics that accurately represent 'intervention effects'; and valid statistical comparisons focusing on effect estimates, confidence intervals and scaled heterogeneity estimates (PIs). However, the search, screening, and analysis strategy are not without limitations.

As with any systematic search strategy, it is possible for primary study titles/abstracts to use alternate or unforeseen wording which can result in otherwise relevant studies failing to appear in the search results. Our decision to omit the word "nature" as a standalone term in the search (and consequent use of proximity operators to link it to with other related terms from the literature) may have increased such a risk. This decision was due to an untenably excessive amount of unrelated material retrieved during the pilot search, driven by the word's polysemous nature. To mitigate the risk associated with this choice, we manually screened any citations with apparent face-validity for inclusion in our review that were referenced in articles that received full-text screening. A further limitation was the possibility that there were excluded articles from grey literature and non-English journals/databases, that may have otherwise met all remaining eligibility criteria targeting experimental rigour. Non-English articles were excluded due to an inability to validly translate such texts, and grey literature was excluded with the aim of enhancing study quality through peer-review. However, future review efforts could apply expanded search strategies and eligibility to avoid the limitations of these decisions.

Additionally, by applying a common denominator design there can substantial nuance lost in comparison to the analyses conducted in primary studies. This is mostly seen in the loss of interactions and the influence of covariates with our natural/non-natural contrasts either collapsing across or separating these additional variables altogether. For example, where Collado and Manrique's (2020) analysis allows for interaction between environement (Natural/Urban) and Awe (Awe/-Mundane), our extraction of their data created two natural-vs-built effects, one for the awe conditions and another for the mundane conditions. We also omitted effects that reflected pre-to-post change from multi-day or weeks-long interventions. While opting against pooling effects from such clearly distinct modes of intervention (relative to the predominating minutes-long exposures) improved the specificity and reliability of the effect estimates, this decision precluded further investigation of other time-related variables, such as frequency of exposure, and how this may interact with each exposure duration. Along with other design elements, such as the categorisation of cognitive domains or intervention characteristics into subgroups, we acknowledge this form of review-author decision making represents an inherent limitation to evidence synthesis.

In hypothesising the effect of exposure duration, we frame intervention duration as a proxy of 'dose' and predicted a dose-response curve (section 1.2). However, this operationalisation may carry an inherent limitation. As previously outlined by Bratman et al. (2019), in medicine/toxicology (where these terms are predominantly applied) there is a distinction between the amount of a substance that merely reaches the organism (i.e., exposure) and the amount of a substance actually absorbed by the organism (i.e., the dose). It is likely that a similar distinction occurs for environmental interventions; even where exposure durations are matched, experimental samples could vary in their engagement with the environment and thus true 'dose'. Experiments included in this review addressed these concepts (e.g., differing engagement due to presence of devices; Jiang et al., 2019). It is also likely that previous experiences in nature influence one's current experience (e.g., Conditioned Restoration; Egner et al., 2020). Overall, the engagement with, and thus *experience of* these experimental nature interventions is a largely unknown variable, and likely interacts with exposure duration to influence restoration. This factor was unable to be accounted for in the current review.

The ability to draw meaningful conclusions from our subgroup analyses within-domains is limited by reducing statistical power with fewer studies/effects, and the inherent observational (non-causal) nature of subgroup analysis. On several occasions, the identical grouping of effects/studies would constitute two different sub-group comparisons; for example, all virtual exposures in the Orienting domain were sedentary, while all actual exposures involved walking, creating a perfect overlap (and thus identical analysis results) for the two different subgroup contrasts. Had a meaningful relationship emerged, it would be impossible to determine which intervention heterogeneity was more relevant – indeed, it is also possible that an unknown factor could be responsible for any observed subgroup correlation.

Future research can increasingly focus on mixed methods to provide greater insight to the qualitative experience of nature interventions, alongside the observation of quantitative restorative effects. Our illustration of the evidence (Fig. 3) provides some insight to lacking experimental observations, such as in clinical and older adult samples – particularly in longer exposure durations – or outcomes measuring vigilance/sustained attention after fatigue induction. Future experiments can address such gaps. Future reviews can synthesise longitudinal studies to expand investigation of how 'amount of exposure' influences effects, including a potential cumulative effect of exposure frequency within such programs. Lastly, a systematic evidence map (e.g., White et al., 2020) could be an ideal method to formally address the vast and heterogenous evidence base regarding nature exposures and various health outcomes outside the scope of the current review.

4.4. Conclusion

In summary, the current review sought to 1) estimate the benefit of natural vs non-natural exposures on cognitive performance for different cognitive domains, 2) assess the relationship between nature benefits and exposure durations, and 3) investigate potential sources of heterogeneity. Our findings indicate that on average, nature interventions are reliably more cognitively restorative than non-natural ones. This effect is also reliably observed in Working Memory and Attentional Control domains, although no differences in domain effects were themselves statistically significant. The average nature benefits are likely to be even more pronounced when the samples underwent a form of experimental fatigue induction prior to the intervention. Furthermore, the overall cognitive difference is significantly moderated by exposure duration,

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whereby nature benefits tend to be greater over short breaks, with a less clear distinction between natural and non-natural exposures when durations near an hour, potentially due to a homeostatic ceiling effect. However, this same relationship was not observed in each cognitive domain, with effects again greatest for Working Memory and Attentional Control.

We conclude that for cognitive performance improvements following acute (<1 h) breaks, nature exposures provide on average a greater benefit for directed attention restoration, particularly from states of cognitive fatigue. These benefits appear maximal after approximately 30-min of exposure, making nature exposures appear particularly suited for maximising short breaks from cognitively demanding tasks – a common circumstance in virtually all modern educational and occupational contexts. We believe these findings may guide any future development of recommendations in practice guidelines for the provision of clinical and/or non-clinical interventions to improve health outcomes relating to cognitive restoration.

CRediT authorship contribution statement

Cameron N. Bell: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualisation. **Rebecca St George:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualisation. **Cynthia Honan:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualisation. **Lachlan J. Bell:** Investigation. **Alex T.W. Jolly:** Investigation. **Allison Matthews:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualisation.

Data availability

All relevant data and analysis codes are accessible at https://osf.io/rb39w/

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvp.2025.102632.

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